Reader beware: I have used local scoping on my claim numbering, i.e. there are numerous claim 1-s in the document. I think this should not be a problem as it is never ambiguous which claim I am refering to.

Question 1. 1. Proposition 3.1b in Humphreys states if I is a solveable ideal of L such that L/I is solveable, then L is solveable. Since $L/[L,L]$ is abelian by definition of $[L,L]$ and $[L,L]$ is nilpotent hence solveable we conclude by the proposition that L is solveable. □

2. We first note that $\mathrm{ad} : L \to \mathfrak{gl}(L)$ is a homomorphism, to see this we employ the Jacobi identity

$$[\mathrm{ad}_x, \mathrm{ad}_y](z) = \mathrm{ad}_x \mathrm{ad}_y(z) - \mathrm{ad}_y \mathrm{ad}_x(z) = [x,[y,z]] - [y,[x,z]] \overset{\text{Jacobi}}{=} [[x,y],z] = \mathrm{ad}_{[x,y]} \tag{1}$$

Thus by Humphreys Proposition 3.1(a) L being solveable implies $\mathrm{ad}(L)$ is a solveable subalgebra of $\mathfrak{gl}(L)$. Thus by Lie's theorem there exists a basis of L such that $\mathrm{ad}(L) \subset \mathfrak{t}(F, \dim L)$ under the identification of $\mathfrak{gl}(L)$ with $\mathfrak{gl}(F, \dim L)$. In part 4 of this problem we show that $[\mathfrak{t},\mathfrak{t}] = \mathfrak{n}$ the subalgebra of strictly upper triangular matrices. Thus it follows that

$$\mathrm{ad}[L,L] = [\mathrm{ad}\, L, \mathrm{ad}\, L] \subset [\mathfrak{t},\mathfrak{t}] = \mathfrak{n}(F, \dim L) \tag{2}$$

Since elements of $\mathfrak{n}(F, \dim L)$ are nilpotent, equation (2) tells us that $[L,L]$ is ad-nilpotent, thus by Engel's theorem $[L,L]$ is nilpotent. □

3. Consider the nonabelian 2 dimensional Lie algebra $L = \mathrm{span}_F(x,y)$ with relation $[x,y] = x$.

$$L^{(1)} = \mathrm{span}_F \{x\}, \quad L^{(2)} \overset{\text{def}}{=} [\mathrm{span}_F \{x\}, \mathrm{span}_F \{x\}] = 0 \tag{3}$$

However the central series has $L^1 = \mathrm{span}_F \{x\}$. It follows by induction that $L^k = [L, L^{k-1}] = [L, \mathrm{span}_F \{x\}] = \mathrm{span}_F(x)$ for all $k > 1$, hence L is not nilpotent. If you don't believe me that L is a lie algebra I give a calculation of the Jacobi identity.

$$[x,[x,y]] + [x,[y,x]] + [y,[x,x]] = [x,x] + [x,-x] + [y,0] = 0 + -[x,x] + 0 = 0 \tag{4}$$
$$[y,[x,y]] + [x,[y,y]] + [y,[y,x]] = [y,x] + [x,0] + [y,-x] = [y,x] + 0 - [y,x] = 0 \tag{5}$$

□

4. Throughout this exercise I will denote $\mathfrak{n}$ as the subset of strictly upper triangular matrices, and for $k \geq 1$ I will denote $\mathfrak{n}_k$ as the subset of matrices $\{\sum a_{ij}e_{ij} \mid a_{ij} \in F, j \geq i+k\}$. I will first compute the lower central series:

Claim 1. $[L,L] \subset \mathfrak{n}$

Claim 2. $[L,\mathfrak{n}] \supset \mathfrak{n}$

Claim 2 tells us in particular that $[L,L] \supset [L,\mathfrak{n}] \supset \mathfrak{n}$, so that using claim 1, $[L,L] = \mathfrak{n}$, it follows by induction and claim 1 that that $L^k = [L, L^{k-1}] = [L,\mathfrak{n}] = \mathfrak{n}$ is the lower central series.

The derived series requires sharper claims:

Claim 3. $[\mathfrak{n}_k, \mathfrak{n}_r] \subset \mathfrak{n}_{k+r}$

Claim 4. $[\mathfrak{n}_k, \mathfrak{n}_r] \supset \mathfrak{n}_{k+r}$

Using claims 3 and 4 and the solution for the lower central series we get the following derived series (note that for $k \geq n$ we have $\mathfrak{n}_k = 0$) $L^0 = \mathfrak{t}$, $L^1 = L^1 = \mathfrak{n}$ then by induction and using claims 3 and 4, we find for $k > 1$ that $L^{(k)} = [\mathfrak{n}_{2^{k-2}}, \mathfrak{n}_{2^{k-2}}] = \mathfrak{n}_{2^{k-1}}$. □

Proof of Claim 1. Let $A = \sum a_{ij}e_{ij}, B = \sum b_{ij}e_{ij} \in L$ and let $C = [A,B] = \sum c_{ij}e_{ij}$, then it suffices to show that $c_{jj} = 0$ for each j, this is straightforward since

$$c_{jj} = \sum_k a_{jk}b_{kj} - b_{jk}a_{kj} \tag{6}$$

but then since $a_{jk} = 0 = b_{jk}$ for $k < j$ and $b_{kj} = a_{kj} = 0$ for $k > j$ we find that

$$c_{jj} = a_{jj}b_{jj} - b_{jj}a_{jj} = 0 \tag{7}$$

□

Proof of Claim 2. It suffices to show that $e_{ij} \in [L, \mathfrak{n}]$ for all $j > i$, so let $j > i$ then $e_{ii} \in L$ and $e_{ij} \in \mathfrak{n}$ so that

$$[e_i i, e_{ij}] = e_{ii}e_{ij} - e_{ij}e_{ii} = e_{ij} \in [L, \mathfrak{n}] \tag{8}$$

□

Proof of Claim 3. Let $A = \sum a_{ij}e_{ij} \in \mathfrak{n}_s$ and $B = \sum a_{ij}e_{ij} \in \mathfrak{n}_r$, it suffices to show that $AB \in \mathfrak{n}_{s+r}$, because then by symmetry te same result will hold for BA and thus also their difference. So let $AB = \sum c_{ij}e_{ij}$, it follows that

$$c_{ij} = \sum_k a_{ik}b_{kj} \overset{A\in\mathfrak{n}_s}{=} \sum_{k\geq i+s} a_{ik}b_{kj} \overset{B\in\mathfrak{n}_r}{=} \sum_{j-r\geq k\geq i+s} a_{ik}b_{kj} \tag{9}$$

Thus the sum is only nonzero when $j \geq i + s + r$ thus $c_{ij} = 0$ for all $j < i + s + r$, this proves that $AB \in \mathfrak{n}_{s+r}$. □

Proof of Claim 4. It suffices to show that $e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r]$ whenever $n \geq j \geq i + s + r$. Choose such an e_{ij}, then $e_{i,j-r} \in \mathfrak{n}_s$ and $e_{j-r,j} \in \mathfrak{n}_r$ so that

$$[e_{i,j-r}, e_{j-r,j}] = e_{i,j-r}e_{j-r,j} - \underbrace{e_{j-r,j}e_{i,j-r}}_{=0 \text{ since } j>i} = e_{ij} \in [\mathfrak{n}_s, \mathfrak{n}_r] \tag{10}$$

□

Question 2. 1. We first compute the matrices of ad_x and ad_y in $\mathfrak{gl}(L)$ with respect to the (ordered) basis $\{x, y\}$, this gives us

$$\operatorname{ad}_x = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \text{ and } \operatorname{ad}_y = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix} \tag{11}$$

It follows that $\operatorname{ad}_x^2 = \operatorname{ad}_x$, $\operatorname{ad}_y^2 = 0$, $\operatorname{ad}_x \operatorname{ad}_y = \operatorname{ad}_y$ and $\operatorname{ad}_y \operatorname{ad}_x = 0$, by taking the trace of these matrices we directly read off the killing form: $\kappa(x, x) = 1$, $\kappa(y, y) = 0$ and $\kappa(x, y) = 0 = \kappa(y, x)$ thus relative to this basis the killing form ahs the following matrix

$$\kappa = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \tag{12}$$

We recall that $\operatorname{Rad}(\kappa) = \{z \in L \mid \kappa(z, -) \equiv 0\}$. Our prior computation shows that $\operatorname{Rad}(\kappa) = \operatorname{span}_F \{y\} = [L, L]$. □

2. Since the identity matrix $1_{n\times n} \notin \mathfrak{sl}(n)$ and $\dim \mathfrak{sl}(n) = n^2 - 1$ (see Humphreys section 1.1), we know that $\operatorname{span}_F \{1_{n\times n}\} \oplus \mathfrak{sl}(n)$ is a subspace of $\mathfrak{gl}(n)$ having dimension $n^2 - 1 + 1 = n^2 = \dim \mathfrak{gl}(n)$, hence is the entire vectorspace. Moreover if $x, y \in \mathfrak{gl}(n)$ symmetry of trace gives us that

$$\operatorname{tr}([x, y]) = \operatorname{tr}(xy) - \operatorname{tr}(yx) = \operatorname{tr}(xy) - \operatorname{tr}(xy) = 0 \tag{13}$$

in particular this holds for $x \in \mathfrak{sl}(n)$ so that $\mathfrak{sl}(n)$ is an ideal. Thus by part 4 of this problem, for $x, y \in \mathfrak{sl}(n)$ we have $\kappa_{\mathfrak{sl}(n)}(x, y) = \kappa_{\mathfrak{gl}(n)}(x, y)$ thus it suffices to compute the latter with respect to the basis $\{e_{ij}\}$ for $\mathfrak{gl}(V)$. Moreover, since κ is bilinear and trace is linear, it will suffice to show the identity holds on the basis $\{e_{ij} \mid i \neq j\} \cup \{e_{ii} - e_{i+1,i+1}\}$ of $\mathfrak{sl}(n)$. Symmetry of κ and trace imply that the following 3 cases are sufficient

Case 1. $x = e_{ij}, y = e_{kl}$*:* We first compute the action of $\mathrm{ad}_x \circ \mathrm{ad}_y$ on each $e_{st} \in \mathfrak{gl}(n)$.

$$[e_{ij}, [e_{k,\ell}, e_{st}]] = [e_{ij}, \delta_{\ell s} e_{kt} - \delta_{tk} e_{s\ell}] = \delta_{jk}\delta_{\ell s} e_{it} - \delta_{js}\delta_{tk} e_{i\ell} + \delta_{\ell i}\delta_{tk} e_{sj} - \delta_{ti}\delta_{\ell s} e_{kj} \tag{14}$$

When we take the trace of $\mathrm{ad}_x \mathrm{ad}_y$ we will sum over s, t the coefficients of the diagonal entries, so that the terms e_{ab} in the above sum are replaced with $\delta_{as}\delta_{bt}$, writing this explicitly we get

$$\begin{aligned}
\mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) &= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \delta_{js}\delta_{tk}\delta_{is}\delta_{\ell t} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \delta_{ti}\delta_{\ell s}\delta_{ks}\delta_{jt} && (15)\\
&= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \delta_{ij}\delta_{tk}\delta_{is}\delta_{\ell t} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \delta_{ti}\delta_{\ell s}\delta_{k\ell}\delta_{jt} && (16)\\
&= \sum_{s,t} \delta_{jk}\delta_{\ell s}\delta_{is} - \underbrace{\delta_{ij}\delta_{tk}\delta_{is}\delta_{\ell t}}_{=0 \text{ since } i\neq j} + \delta_{\ell i}\delta_{tk}\delta_{jt} - \underbrace{\delta_{ti}\delta_{\ell s}\delta_{k\ell}\delta_{jt}}_{=0 \text{ since } k\neq \ell} && (17)\\
&= \sum_{s,t} \delta_{jk}\delta_{i\ell}\delta_{\ell s} + \sum_{s,t} \delta_{jk}\delta_{i\ell}\delta_{jt} && (18)\\
&= \sum_{\substack{1\leq t\leq n\\ s=\ell}} \delta_{jk}\delta_{i\ell} + \sum_{\substack{1\leq s\leq n\\ t=j}} \delta_{jk}\delta_{i\ell} && (19)\\
&= 2n\delta_{jk}\delta_{i\ell} && (20)\\
&= 2n\,\mathrm{tr}(e_{ij}e_{k\ell}) && (21)\\
&= 2n\,\mathrm{tr}(xy) && (22)
\end{aligned}$$

Case 2. $x = e_{ii} - e_{i+1,i+1}, y = e_{k\ell}$*:* Similarly to the above we first compute the action of $\mathrm{ad}_x \circ \mathrm{ad}_y$ on each e_{st}.

$$\begin{aligned}
[e_{ii} - e_{i+1,i+1}, [e_{k\ell}, e_{s,t}]] &= [e_{ii} - e_{i+1,i+1}, \delta_{\ell s} e_{kt} - \delta_{tk} e_{s\ell}] && (23)\\
&= \delta_{\ell s}\delta_{ik} e_{it} - \delta_{is}\delta_{tk} e_{i\ell} - \delta_{i+1,k}\delta_{\ell s} e_{i+1,t} + \delta_{i+1,s}\delta_{tk} e_{i+1,\ell} && (24)\\
&\quad - \delta_{ti}\delta_{\ell s} e_{ki} + \delta_{i\ell}\delta_{tk} e_{si} + \delta_{t,i+1}\delta_{\ell s} e_{k,i+1} - \delta_{\ell,i+1}\delta_{tk} e_{s,i+1} && (25)
\end{aligned}$$

We use the same reasoning as case 1 to compute the trace being

$$\begin{aligned}
\mathrm{tr}(\mathrm{ad}_x \mathrm{ad}_y) =& \sum_{s,t} \delta_{\ell s}\delta_{ik}\delta_{is} - \delta_{is}\delta_{tk}\delta_{is}\delta_{\ell t} - \delta_{i+1,k}\delta_{\ell s}\delta_{i+1,s} + \delta_{i+1,s}\delta_{tk}\delta_{i+1,s}\delta_{\ell t} && (26)\\
&+ \sum_{s,t} -\delta_{ti}\delta_{\ell s}\delta_{ks}\delta_{it} + \delta_{i\ell}\delta_{tk}\delta_{it} + \delta_{t,i+1}\delta_{\ell s}\delta_{k,s}\delta_{i+1,t} - \delta_{\ell,i+1}\delta_{tk}\delta_{i+1,t} && (27)\\
=& \sum_{s,t} \underbrace{\delta_{\ell s}\delta_{ik}\delta_{is}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{tk}\delta_{is}\delta_{\ell t}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{i+1,k}\delta_{\ell s}\delta_{i+1,s}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{i+1,s}\delta_{tk}\delta_{\ell,t}}_{0 \text{ since } k\neq \ell} && (28)\\
&+ \sum_{s,t} - \underbrace{\delta_{\ell s}\delta_{ks}\delta_{it}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{i\ell}\delta_{tk}\delta_{it}}_{0 \text{ since } \ell\neq k} + \underbrace{\delta_{\ell s}\delta_{ks}\delta_{i+1,t}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{\ell,i+1}\delta_{tk}\delta_{i+1,t}}_{0 \text{ since } \ell\neq k} && (29)\\
=& 0 && (30)\\
=& 2n\Big(\underbrace{\delta_{i\ell}\delta_{ik}}_{0 \text{ since } \ell\neq k} - \underbrace{\delta_{i+1,\ell}\delta_{i+1,k}}_{0 \text{ since } \ell\neq k}\Big) && (31)\\
=& 2n\,\mathrm{tr}((e_{ii} - e_{i+1,i+1})e_{k\ell}) && (32)\\
=& 2n\,\mathrm{tr}(xy) && (33)
\end{aligned}$$

Case 3. $x = e_{ii} - e_{i+1,i+1}, y = e_{jj} - e_{j+1,j+1}$*:* TODO

3. By theorem 5.1, it suffices to show that the killing form is non-degenerate, i.e. every $x \in \mathfrak{sl}(n)$ has some $y \in \mathfrak{sl}(n)$ such that $\kappa(x, y) = 2n\,\mathrm{tr}(xy) \neq 0$, write $x = \sum x_{ij}e_{ij}$, then if $x_{ij} \neq 0$ for some

$i \neq j$, then

$$xe_{ji} = \sum_{k,\ell} x_{k,\ell} e_{k\ell} e_{ji} = \sum_{k} x_{kj} e_{ki} \tag{34}$$

So that $\operatorname{tr}(xe_{ji}) = x_{ij} \neq 0$. Otherwise we can assume x is diagonal, then since $x \neq 0$ there is some diagonal entry $x_{ii} \neq 0$, furthermore since $\operatorname{tr}(x) = 0$ there must be some other diagonal entry x_{jj} with $x_{jj} \neq x_{ii}$. It follows that

$$x(e_{ii} - e_{jj}) = x_{ii}e_{ii} - x_{jj}e_{jj} \tag{35}$$

and $\operatorname{tr}(x(e_{ii} - e_{jj})) = x_{ii} - x_{jj} \neq 0$. Thus in either case we can choose $y = e_{ji}$ or $y = (e_{ii} - e_{jj})$ so that $\operatorname{tr}(xy) \neq 0$. Now applying the previous subpart we find that

$$\kappa(x, y) = 2n \operatorname{tr}(xy) \neq 0 \tag{36}$$

We conclude that the killing form is nondegenerate and thus $\mathfrak{sl}(n)$ is semisimple by Humphreys Theorem 5.1. □

4. Since $I \subset L$ is a subspace, it has a compliment S satisfying $L = I \oplus S$ as a vector space. We may take $v_1, \ldots, v_r$ to be a basis for I and $v_{r+1}, \ldots, v_n$ to be a basis for S. Whence any element of $\mathfrak{gl}(L)$ can be written in the form $\sum a_{ij}e_{ij}$ for $a_{ij} \in F$ and $e_{ij}(v_k) = \delta_{jk}v_i$. Now let $x, y \in I$, since I is an ideal we know that $\operatorname{ad}_x, \operatorname{ad}_y : L \to I$, so in particular so does their compositon. In terms of our basis this tells us that for $\operatorname{ad}_x = \sum x_{ij}e_{ij}, \operatorname{ad}_y = \sum y_{ij}e_{ij}$ and $\operatorname{ad}_x \circ \operatorname{ad}_y = \sum a_{ij}e_{ij}$ we have $x_{ij} = y_{ij} = a_{ij} = 0$ for $i > r$, whence in particular $e_{ii} = 0$ for $i = r+1, \ldots, n$ it follows that

$$\kappa(x, y) = \operatorname{tr}(\operatorname{ad}_x \circ \operatorname{ad}_y) = \sum_{1}^{n} a_{ii} = \sum_{1}^{r} a_{ii} \tag{37}$$

$$= \sum_{i=1}^{r}\sum_{k=1}^{n} x_{ik}y_{ki} = \sum_{i=1}^{r}\sum_{k=1}^{r} x_{ik}y_{ki} = \operatorname{tr}(\operatorname{ad}_x|_I \circ \operatorname{ad}_y|_I) = \kappa_I(x, y) \tag{38}$$

So for any $(x, y) \in I \times I$ we have $\kappa(x, y) = \kappa_I(x, y)$, this of course implies that $\kappa|_{I\times I} = \kappa_I$. □

Now to see that this result does not hold for subalgebras consider the nonabelian subalgebra of dimension 2 given in teh first part of the question, and consider the subalgebra $H = \operatorname{span}_F\{x\}$, we computed in part 1 that $\kappa(x, x) = 1$ whence $\kappa|_{H\times H}$ has matrix 1, but since H is abelian ad_x acts as zero on H so that $\kappa_H(x, x) = 0$, so that κ_H has matrix 0. □

Question 3. 1. Since toral subalgebras are abelian (Humphreys Lemma 8.1) it suffices to show that H is maximal abelian. Using the computation from Humphreys section 1.2 we know that matrices in $\mathfrak{sp}(4)$ are of the following form

$$\begin{pmatrix} A & B \\ C & -A^t \end{pmatrix} \tag{39}$$

where $A, B, C \in \mathfrak{gl}(2)$ and B, C are symmetric. Now we take $x = \operatorname{Diag}(1, 1, -1, -1) \in H$ and compute

$$\left[x, \begin{pmatrix} A & B \\ C & -A^t \end{pmatrix}\right] = \begin{pmatrix} 0 & 2B \\ -2C & 0 \end{pmatrix} \tag{40}$$

so that an abelian subalgebra containing H must only contain matrices of the form $(A, 0; 0, -A^t)$. We will be done once we show that A need be diagonal for such an element to be included in H. So

choose a non-diagonal matrix $A = (a, b; c, d)$ for b, c not both zero, then $x = \mathrm{Diag}(1, -1, -1, 1) \in H$ and

$$\left[x, \begin{pmatrix} A & 0 \\ 0 & -A^t \end{pmatrix}\right] = \begin{pmatrix} \begin{pmatrix} 0 & 2b \\ -2c & 0 \end{pmatrix} & 0 \\ 0 & \begin{pmatrix} 0 & 2c \\ -2b & 0 \end{pmatrix} \end{pmatrix} \neq 0 \tag{41}$$

It follows that any abelian subalgebra containing H contains only diagonal matrices, but H already contains all the diagonal matrices in $\mathfrak{sp}(2)$ this shows that H is maximal abelian and thus maximal toral by Humphreys Lemma 8.1. □

2. We take the basis $h_1 = e_{11} - e_{33}$ and $h_2 = e_{22} - e_{44}$ for H. The roots space decomposition is as follows:

$$L_{\beta_1} = \mathrm{span}_F \{e_{12} - e_{43}\}, \beta_1(h_1) = 1, \beta_1(h_2) = -1 \tag{42}$$
$$L_{\beta_2} = \mathrm{span}_F \{e_{21} - e_{34}\}, \beta_2(h_1) = -1, \beta_2(h_2) = 1 \text{ so that } \beta_2 = -\beta_1 \tag{43}$$
$$L_{\beta_3} = \mathrm{span}_F \{e_{14} + e_{23}\}, \beta_3(h_1) = 1, \beta_3(h_2) = 1 \tag{44}$$
$$L_{\beta_4} = \mathrm{span}_F \{e_{32} + e_{41}\}, \beta_4(h_1) = -1, \beta_4(h_2) = -1 \text{ so that } \beta_4 = -\beta_3 \tag{45}$$
$$L_{\beta_5} = \mathrm{span}_F \{e_{13}\}, \beta_5(h_1) = 2, \beta_5(h_2) = 0 \tag{46}$$
$$L_{\beta_6} = \mathrm{span}_F \{e_{31}\}, \beta_6(h_1) = -2, \beta_6(h_2) = 0 \text{ so that } \beta_6 = -\beta_5 \tag{47}$$
$$L_{\beta_7} = \mathrm{span}_F \{e_{24}\}, \beta_7(h_1) = 0, \beta_7(h_2) = 2 \tag{48}$$
$$L_{\beta_8} = \mathrm{span}_F \{e_{42}\}, \beta_8(h_1) = 0, \beta_8(h_2) = -2 \text{ so that } \beta_8 = -\beta_7 \tag{49}$$

Writing explicitly the roots as linear functionals let $A = \mathrm{Diag}(a, b, -a, -b) \in H$, then

$$\beta_1(A) = a - b = -\beta_2(A) \tag{50}$$
$$\beta_3(A) = a + b = -\beta_4(A) \tag{51}$$
$$\beta_5(A) = 2a = -\beta_6(A) \tag{52}$$
$$\beta_7(A) = 2b = -\beta_8(A) \tag{53}$$

□

3. Take $\alpha_1 = \beta_1$ in the previous subpart and $\alpha_2 = \beta_7$, this lets us rewrite the root system as

$$\Phi = \{\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2), \pm(2\alpha_1 + \alpha_2)\} \tag{54}$$

So that indeed $\Delta = \{\alpha_1, \alpha_2\}$ is a base for the root system by definition (they form a basis for the root system and all other elements are strictly positive or negative integer linear combinations of them) so that these are the simple roots. To compute the Cartan matrix we use the computations of the elements t_α from the fourth subpart, as these are needed to take inner products. Now, the definition of (α, β) for $\alpha, \beta \in \Phi$ is given in section 8.5 of Humphreys as $(\alpha, \beta) = \kappa(t_\alpha, t_\beta)$, and moreover $\langle \alpha, \beta \rangle := \frac{2(\alpha,\beta)}{(\alpha,\alpha)}$ in Humphreys 9.1. The Cartan matrix as defined in Humphreys 11.1 is the matrix $\sum_{i,j} \langle \alpha_i, \alpha_j \rangle e_{ij}$ summing over the simple roots. Since we computed $t_{\alpha_1}, t_{\alpha_2}$ in the subsequent subpart of the question we can use Claim 1 (from part 4 of this question) to compute (α_i, α_j) for $(i, j) \in \{1, 2\}^2$. In fact we already computed the case $i = j$ when computing the coroots.

$$(\alpha_1, \alpha_1) \overset{(81)}{=} \frac{1}{6} \tag{55}$$
$$(\alpha_2, \alpha_2) \overset{(82)}{=} \frac{1}{3} \tag{56}$$
$$(\alpha_2, \alpha_1) = (\alpha_1, \alpha_2) = 6\,\mathrm{tr}(t_{\alpha_1}, t_{\alpha_2}) = -\frac{1}{12} \tag{57}$$
$$\tag{58}$$

We plug these in to compute $\langle \alpha_i, \alpha_j \rangle$, where it is immediate from the definition that when $i = j$ this has value 2, so in the case of $i \neq j$:

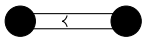
$$\langle \alpha_1, \alpha_2 \rangle = -\frac{2}{6}/\frac{1}{6} = -2 \tag{59}$$

$$\langle \alpha_2, \alpha_1 \rangle = -\frac{2}{6}/\frac{1}{3} = -1 \tag{60}$$

Thus we compute the Cartan matrix as:

$$\begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix} \tag{61}$$

Now it is trivial to compute the Dynkin diagram of Φ by using the procedure described in Humphreys 11.2, since $\dim \operatorname{span}_F \Phi = 2$ the number of simple roots we start with two vertices, and draw $\langle \alpha_1, \alpha_2 \rangle \langle \alpha_2, \alpha_1 \rangle = (-1)(-2) = 2$ lines between them. Now all that remains is to draw an arrow towards the shorter root, in this case $(\alpha_1, \alpha_1) = \frac{1}{6} < \frac{1}{3} = (\alpha_2, \alpha_2)$ so the arrow is from α_2 to α_1, the diagram given from this procedure is drawn below (I am using the Dynkin Diagrams tikz package):



(62)

Which is of type C2. □

4. Humphreys proposition 8.3 says that for $x \in L_\alpha$, $y \in L_{-\alpha}$ we have $[x, y] = \kappa(x, y) t_\alpha$. We need the following claims:

Claim 1. If $h, h' \in H$ then $\kappa(h, h') = 6 \operatorname{tr}(hh')$

Claim 2. Let $\alpha \in \Phi$ then take root vectors $x_\alpha \in L_\alpha$, $y_\alpha \in L_{-\alpha}$ and $h \in \Phi$, then

$$\kappa(x_\alpha, y_\alpha) = \frac{6 \operatorname{tr}(h[x_\alpha, y_\alpha])}{\alpha(h)} \tag{63}$$

Using the claims we find that for $\alpha \in \Phi$

$$t_\alpha = \frac{[x_\alpha, y_\alpha]}{\kappa(x_\alpha, y_\alpha)} = \frac{\alpha(h)[x_\alpha, y_\alpha]}{6 \operatorname{tr}(h[x_\alpha, y_\alpha])} \tag{64}$$

Thus we compute each $[x_\alpha, y_\alpha]$, $h[x_\alpha, y_\alpha]$ and $\alpha(h)$ where $h \in \{h_1, h_2\}$ so that $\alpha(h) \neq 0$ below:

$$[x_{\alpha_1}, y_{\alpha_1}] = [e_{12} - e_{43}, e_{21} - e_{34}] = e_{11} + e_{44} - e_{22} - e_{33} = h_1 - h_2 \tag{65}$$

$$[x_{\alpha_2}, y_{\alpha_2}] = [e_{24}, e_{42}] = e_{22} - e_{44} = h_2 \tag{66}$$

$$[x_{\alpha_1+\alpha_2}, y_{\alpha_1+\alpha_2}] = [e_{14} + e_{23}, e_{32} + e_{41}] = e_{11} + e_{22} - e_{33} - e_{44} = h_1 + h_2 \tag{67}$$

$$[x_{2\alpha_1+\alpha_2}, y_{2\alpha_1+\alpha_2}] = [e_{13}, e_{31}] = e_{11} - e_{33} = h_1 \tag{68}$$

$$h_1[x_{\alpha_1}, y_{\alpha_1}] = e_{11} + e_{33} \tag{69}$$

$$h_2[x_{\alpha_2}, y_{\alpha_2}] = e_{22} + e_{44} \tag{70}$$

$$h_1[x_{\alpha_1+\alpha_2}, y_{\alpha_1+\alpha_2}] = e_{11} + e_{33} \tag{71}$$

$$h_1[x_{2\alpha_1+\alpha_2}, y_{2\alpha_1+\alpha_2}] = e_{11} + e_{33} \tag{72}$$

$$\alpha_1(h_1) = 1 \tag{73}$$

$$\alpha_2(h_2) = 2 \tag{74}$$

$$(\alpha_1 + \alpha_2)(h_1) = 1 \tag{75}$$

$$(2\alpha_1 + \alpha_2)(h_1) = 2 \tag{76}$$

Whence we use the formula from equation (64) to compute each t_α (note we only do the positive integer combinations since $t_{-\alpha} = -t_\alpha$, this is readily seen as $\kappa(t_{-\alpha}, h) \stackrel{\text{def}}{=} -\alpha(h) \stackrel{\text{def}}{=} -\kappa(t_\alpha, h) = \kappa(-t_\alpha, h)$):

$$t_{\alpha_1} = \frac{\alpha_1(h_1)(h_1 - h_2)}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{12}(h_1 - h_2) \tag{77}$$

$$t_{\alpha_2} = \frac{\alpha_2(h_2)h_2}{6\,\mathrm{tr}(e_{22} + e_{44})} = \frac{1}{6}h_2 \tag{78}$$

$$t_{\alpha_1+\alpha_2} = \frac{(\alpha_1 + \alpha_2)(h_1)(h_1 + h_2)}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{12}(h_1 + h_2) \tag{79}$$

$$t_{2\alpha_1+\alpha_2} = \frac{(2\alpha_1 + \alpha_2)(h_1)h_1}{6\,\mathrm{tr}(e_{11} + e_{33})} = \frac{1}{6}h_1 \tag{80}$$

Now to compute the coroots, we use Claim 1 to compute $\kappa(t_\alpha, t_\alpha)$ for each α.

$$\kappa(t_{\alpha_1}, t_{\alpha_1}) = 24/12^2 = \frac{1}{6} \tag{81}$$

$$\kappa(t_{\alpha_2}, t_{\alpha_2}) = \frac{12}{6^2} = \frac{1}{3} \tag{82}$$

$$\kappa(t_{\alpha_1+\alpha_2}, t_{\alpha_1+\alpha_2}) = \frac{24}{12^2} = \frac{1}{6} \tag{83}$$

$$\kappa(t_{2\alpha_1+\alpha_2}, t_{2\alpha_1+\alpha_2}) = 12/6^2 = \frac{1}{3} \tag{84}$$

Whence we compute the h_α from the given formula, since $t_{-\alpha} = -t_\alpha$ bilinearity of the killing form ensures that $\kappa_{-\alpha} = -\kappa_\alpha$, so once again we only compute the positive linear combinations.

$$h_{\alpha_1} = h_1 - h_2 \tag{85}$$

$$h_{\alpha_2} = h_2 \tag{86}$$

$$h_{\alpha_1+\alpha_2} = h_1 + h_2 \tag{87}$$

$$h_{2\alpha_1+\alpha_2} = h_1 \tag{88}$$

The simple coroots are defined in Humphreys 9.2 as

$$\check{\alpha}_1 = \frac{2\alpha_1}{(\alpha_1, \alpha_1)} = 12\alpha_1 \tag{89}$$

$$\check{\alpha}_2 = \frac{2\alpha_2}{(\alpha_2, \alpha_2)} = 6\alpha_2 \tag{90}$$

To classify the coroot system, we simply compute its cartan matrix who's entries are given by $\langle \check{\alpha}_i, \check{\alpha}_j \rangle$, once again it is immediate that for $i = j$ we get 2, so we compute the other two cases:

$$\langle \check{\alpha}_1, \check{\alpha}_2 \rangle = \frac{2(12\alpha_1, 6\alpha_2)}{(12\alpha_1, 12\alpha_1)} = \frac{1}{2}\langle \alpha_1, \alpha_2 \rangle = -1 \tag{91}$$

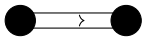
$$\langle \check{\alpha}_2, \check{\alpha}_1 \rangle = \frac{2(6\alpha_2, 12\alpha_1)}{(6\alpha_2, 6\alpha_2)} = 2\langle \alpha_1, \alpha_2 \rangle = -2 \tag{92}$$

Thus we get cartan matrix for $\check{\Phi}$:

$$\begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix} \tag{93}$$

Then the product of the off diagonal entries is the same as the original cartan matrix, and $\dim \mathrm{span}_F(\check{\Phi}) = 2$ so we get the same Dynkin diagram appart from $(\check{\alpha}_1, \check{\alpha}_1) = 12^2(\alpha_1, \alpha_1) =$

$24 > 12 = 6^2(\alpha_2, \alpha_2) = (\check{\alpha}_2, \check{\alpha}_2)$ i.e. the arrow is reversed. Thus the Dynkin diagram is



(94)

Which is of type B2. □

TODO: Prove the claims

5. By Humphrey's theorem 14.2 (and the preceding discussion) an isomorphism of root systems induces an isomorphism of simple lie algebras, in order to apply this we first show that $\mathfrak{sp}(4)$ and $\mathfrak{so}(5)$ are simple, which reduces the argument to showing that $\Phi_{\mathfrak{sp}(4)}$ can be taken to have type C2, then showing that any two root systems of type C2 are isomorphic. This reduces the argument to the following sequence of claims.

   Claim 1. If L is a semisimple lie algebra with irreducible root system Φ_L, then L is simple.

   Claim 2. $\Phi_{\mathfrak{sp}(4)}$ can be taken to have root system of type B2.

   Claim 3. Any two root systems of type B2 are isomorphic.

   Claim 4. Root systems of type B2 and C2 are irreducible.

   Now I will formally prove the result assuming the claims. By claim 4, $\mathfrak{sp}(4)$ and $\mathfrak{so}(5)$ have irreducible root systems, and thus by Claim 1 they are simple. Then claim 2 tells us that $\Phi_{\mathfrak{sp}(4)}$ has an ordered basis of simple roots so that it is of type B2, then claim 3 tells us that $\Phi_{\mathfrak{sp}(4)} \cong \Phi_{\mathfrak{so}(5)}$. Whence $\mathfrak{sp}(4)$ and $\mathfrak{so}(5)$ are simple Lie algebras with isomorphic root systems, so by Humphreys theorem 14.2 and the preceding discussion there is an induced isomorphism $\mathfrak{sp}(4) \cong \mathfrak{so}(5)$. □

   Proof of Claim 1. This is trivially implied by Humphreys corollary 14.1. □

   Proof of Claim 2. Take $\Phi_{\mathfrak{sp}(4)}$ with the ordered basis of simple roots $\Delta = \{\alpha_2, \alpha_1\}$ as defined in the previous 2 subparts. Then the Cartan matrix with this basis is the transpose of the cartan matrix in the ordered basis $\{\alpha_1, \alpha_2\}$, thus agrees with the cartan matrix in equation (93), we saw in the paragraph preceding equation (94) that such a Cartan matrix gives rise to a root system of type B2. □

   Proof of Claim 3. Humphreys proposition 11.1 states that the Cartan matrix specifies a root system up to isomorphism. So it suffices to show that any two root systems of type B2 have the same Cartan matrix. Let Φ be a root system of type B2, then by the algorithm prescribed to compute the Dynkin diagram we know that Φ has two simple roots α, β, forming a basis $\Delta = \{\alpha, \beta\}$, that $\langle\alpha, \beta\rangle\langle\beta, \alpha\rangle = 2$ and that $(\beta, \beta) < (\alpha, \alpha)$. This inequality condition implies that

$$|\langle\alpha, \beta\rangle| = \frac{|(\alpha, \beta)|}{(\alpha, \alpha)} < \frac{|(\alpha, \beta)|}{(\beta, \beta)} = |\langle\beta, \alpha\rangle| \tag{95}$$

   since the only possible integer factorizations of 2 are $\{\pm 1, \pm 2\}$ we get Cartan matrices

$$\begin{pmatrix} 2 & \pm 1 \\ \pm 2 & 2 \end{pmatrix} \tag{96}$$

   Notice that either sign convention of the Cartan matrix specifies the same root system, since the other sign convention simply corresponds to basis $\Delta = \{-\alpha, -\beta\}$. Thus all Lie algebras of type B2 can have basis chosen such that they have the same Cartan matrix which implies they are isomorphic by Humphreys proposition 11.1. □

   Proof of Claim 4. Suppose Φ has type B2 or C2, then by reading off the corresponding Dynkin diagrams (equations (62) and (94)) we conclude that Φ has two simple roots α and β and that $\langle\alpha, \beta\rangle\langle\beta, \alpha\rangle = 2$ so that $(\alpha, \beta) \neq 0$ so that $\Delta = \{\alpha, \beta\}$ is not irreducible. By Humphreys 10.4 (see first paragraph) Φ is irreducible if and only if Δ is. We conclude that Φ is irreducible. □

Question 4. (a) By the universal property of U, there is a unique algebra homomorphism ψ such that the following commutes:

$$\begin{array}{ccc} U & \overset{\exists!\psi}{\dashrightarrow} & U' \\ {\scriptstyle\iota}\uparrow & \nearrow_{\iota'} & \\ L & & \end{array} \tag{97}$$

I.e. a unique algebra homomorphism ψ such that $\psi \circ \iota = \iota'$.

I suspect the question may have intended to ask that U is uniquely isomorphic to U', this is useful for disambiguation so I will prove it here using the following diagram:

$$\begin{array}{ccccc} & & \overset{\exists!}{\frown} & & \\ U & \xrightarrow{\Psi} & U' & \xrightarrow{\Psi'} & U \\ & {\scriptstyle\iota}\nwarrow & {\scriptstyle\iota'}\uparrow & \nearrow{\scriptstyle\iota} & \\ & & L & & \end{array} \tag{98}$$

Then notice that 1_U could be the map on the dashed line but also $\Psi' \circ \Psi$ so that $\Psi' \circ \Psi = 1_U$. The argument for $\Psi \circ \Psi' = 1_{U'}$ is the same up to relabelling. □

(b) Let A be an algebra and $j : L \to A$ be a linear map satisfying the bracket identity. It will suffice to show there exists a unique map $\varphi : T(L)/J \to A$ satisfying $\varphi \circ \iota = j$. By the universal property of the tensor algebra, for any linear map $j : L \to A$ there is a unique algebra morphism $f : T(L) \to A$ satisfying $f \circ s = j$ where $s : L \hookrightarrow T(L)$ is the inclusion. Moreover if $f : T(L) \to A$ satisfies $J \subset \ker f$, then by the first isomorphism theorem there exists a unique algebra morphism $\varphi : T(L)/J \to A$ such that $\varphi \circ \pi = f$ where π is the quotient. Note that since $L \hookrightarrow T(L)$ is injective and $J \subset \ker j$ by assumption, we have $J \subset \ker f$. So the isomorphism theorem does indeed apply. This paragraph can be summarized in the following diagram.

$$\begin{array}{ccccc} L & \xrightarrow{s} & T(L) & \xrightarrow{\pi} & T(L)/J \\ & {\scriptstyle j}\searrow & {\scriptstyle\exists!f}\downarrow & \swarrow{\scriptstyle\exists!\varphi} & \\ & & A & & \end{array} \tag{99}$$

Where f is induced by the universal property of $(T(L), s)$ and φ is induced by the first isomorphism theorem on f since $j(x \otimes y - y \otimes x - [x, y]) = 0$ for all x, y and ι is injective so that $J \subset \ker f$. To see that $\iota(L)$ generates $U(L)$ as an algebra it suffices to show by surjectivity of π that $s(L)$ generates $T(L)$. We prove this by showing that $(s(L)) \subset T(L)$ satisfies the universal propery of $T(L)$. So let A be an algebra and $j : L \to A$ be linear, moreover let e be the inclusion $(s(L)) \hookrightarrow T(L)$, then the universal property of $T(L)$ gives a unique algebra homomorphism $f : T(L) \to A$ with $f \circ s = j$, and by definition the restriction $f|_{(s(L))}$ is the unique algebra homomorphism satisfying $f = f|_{s(L)} \circ e$, notice now that the restriction satisfies $f|_{s(L)} s(x) = f(s(x))$ for all $x \in L$. So that $((s(L)), s)$ satisfies the universal property of $T(L)$. Written diagramatically the argument is

$$\begin{array}{ccccc} L & \xrightarrow{s} & (s(L)) & \xrightarrow{e} & T(L) \\ & {\scriptstyle j}\searrow & & \swarrow{\scriptstyle\exists!f} & \\ & & A & & \end{array} \tag{100}$$

So that the restriction satisfies

$$\begin{array}{ccccc} L & \xrightarrow{s} & (s(L)) & \xrightarrow{e} & T(L) \\ & {\scriptstyle j}\searrow & {\scriptstyle\exists!f|_{(s(L))}}\downarrow & \swarrow{\scriptstyle\exists!f} & \\ & & A & & \end{array} \tag{101}$$

Which is the universal property for $T(L)$.

Now we show that if $L = \text{span}_F \{x_1, \ldots, x_n\}$ with trivial Lie bracket $U(L) = F[x_1, \ldots, x_n]$, thus we show that $(F[x_1, \ldots, x_n], \iota : x_j \mapsto x_j)$ satisfies the universal property. In this case a linear map satisfying the bracket identity is $j : L \to A$ with $j(x)j(y) - j(y)j(x) = 0$ it is immediate that ι satisfies this identity since $F[x_1, \ldots, x_n]$ is abelian. The universal property of the ring $R[x_1, \ldots, x_n]$ is for any ring homomorphism $\phi : R \to S$ and any $s_j \in S$ there is a unique map $\hat{\phi}$ with $\hat{\phi}|_R = \phi$ satisfying $\hat{\phi}(x_j) = s_j$. Any F-algebra homomorphism restricts to $1 : F \to F$, so that for any $\{a_j\}_1^n \subset A$ there exists a unique algebra homomorphism $\phi : x_j \to a_j$. A linear map $L \to A$ is defined by its value on the basis x_j, i.e. is specified by $j(x_j)$, thus by the universal property of $F[x_1, \ldots, x_n]$ there exists a unique map $\phi : F[x_1, \ldots, x_n]$ with $\phi \circ \iota(x_j) = f(x_j) = j(x_j)$ thus a unique ϕ satisfying $\phi \circ \iota = j$. Again we can write this diagramatically.

$$\begin{array}{ccc} L & \xrightarrow{\iota : x_j \mapsto x_j} & F[x_1, \ldots, x_n] \\ \Big\downarrow{\scriptstyle j : x_j \mapsto a_j} & \swarrow_{\exists! \phi : x_j \mapsto a_j} & \\ A & & \end{array} \tag{102}$$

Linearity of j (and ι and ϕ) ensures the diagram commutes just by specifying these a_j values. □

(c) Notice that an L-module structure on V defines a linear map $j : L \to \text{End}(V)$ (*I found it immensely satisfying to notice this*) via $j(x) \mapsto (v \mapsto x.v)$ linearity is implied by M1,M2 and the relation $j([x, y]) = j(x)j(y) - j(y)j(x)$ by M3 (Mj defined as in Humphreys 6.1). Thus the universal property of $U(L)$ gives us

$$\begin{array}{ccc} L & \xrightarrow{\iota} & U(L) \\ \Big\downarrow{\scriptstyle j} & \swarrow_{\exists! \phi} & \\ \text{End}(V) & & \end{array} \tag{103}$$

And a unital $U(L)$ module structure is equivalent to a F-algebra homomorphism $U(L) \to \text{End}(V)$. The uniqueness thus implies the correspondence

$$\hom_{\text{Vect}_F}(L, \text{End}(V)) \cap \text{Image has bracket relation} \to \hom_{F\text{-alg}}(U(L), \text{End}(V)) \tag{104}$$

Via taking the universal property map is injective, so demonstrating surjectivity will prove the equivalence. Consider $\phi \in \hom_{F\text{-alg}}(U(L), \text{End}(V))$, then $\phi \circ \iota$ has the bracket relation by definition of ι, is linear since ϕ and ι are and trivially satisfies commutativity of the following diagram:

$$\begin{array}{ccc} L & \xrightarrow{\iota} & U(L) \\ \Big\downarrow{\scriptstyle \phi \circ \iota} & \swarrow_{\phi} & \\ \text{End}(V) & & \end{array} \tag{105}$$

Thus ϕ is the image of $\phi \circ \iota$ under the universal property map which proves surjectivity. It is trivial to see this maps submodules to submodules since in such a case $W \subset V$ a submnodule the above construction carries through identically by simply replacing each occurance of $\text{End}\, V$ with $\text{End}\, W$. Finally we check that this correspondence matches L-module homs with $U(L)$ module homs. Consider two L-modules V, W, equivalently two linear maps $j_V : L \to \text{End}\, V$ and $j_W : L \to \text{End}(W)$ satisfying the bracket relation. Then an L-module homomorphism is equivalent to an algebra homomorphism Φ such that the following commutes:

$$\begin{array}{ccc} \text{End}\, V & \xrightarrow{\Phi} & \text{End}\, W \\ & {\scriptstyle j_V} \nwarrow \quad \nearrow {\scriptstyle j_W} & \\ & L & \end{array} \tag{106}$$

Applying the correspondence (the universal property) to j_V and j_W gives us F-algebra homomorphisms

ϕ_V and ϕ_W such that the following commutes

$$\begin{array}{ccccc} \operatorname{End} V & & \xrightarrow{\Phi} & & \operatorname{End} W \\ {\scriptstyle j_V}\uparrow & \nwarrow^{\phi_V} & & \nearrow_{\phi_W} & \uparrow{\scriptstyle j_W} \\ L & \xrightarrow{\iota} & U(L) & \xleftarrow{\iota} & L \end{array} \tag{107}$$

We notice that this is exactly a module homomorphism between the $U(L)$ module structure given by ϕ_V and the structure given by ϕ_W. For the reverse direction of the correspondence we start with some $U(L)$ module structures ϕ_V and ϕ_W and an algebra morphism $\Phi : \operatorname{End} V \to \operatorname{End} W$ such that the following commutes:

$$\begin{array}{ccc} \operatorname{End} V & \xrightarrow{\Phi} & \operatorname{End} W \\ & {\scriptstyle \phi_V}\nwarrow \quad \nearrow{\scriptstyle \phi_W} & \\ & U(L) & \end{array} \tag{108}$$

then notice the following commutes thus sending Φ to the corresponding L-module morphism

$$\begin{array}{ccc} \operatorname{End} V & \xrightarrow{\Phi} & \operatorname{End} W \\ & {\scriptstyle \phi_V}\nwarrow \quad \nearrow{\scriptstyle \phi_W} & \\ & U(L) \quad {\scriptstyle \phi_W \circ \iota}\nearrow & \\ & {\scriptstyle \phi_V \circ \iota}\nwarrow \quad \uparrow{\scriptstyle \iota} & \\ & L & \end{array} \tag{109}$$

□

(d) The following claim is essential for many of the arguments used in this subpart:

Claim. If $\sigma \in S_n$ then there is some v which is a sum of (PBW) basis elements involving at most $n-1$ terms such that

$$x_1^{a_1} \cdots x_n^{a_n} = \prod_1^n x_{\sigma(j)}^{a_{\sigma(j)}} + v \tag{110}$$

1. Since $x_1, \ldots, x_n$ are n distinct basis elements in $U(L)$ (by PBW) and are all in the image of ι we conclude that $\operatorname{rank} \iota \geq n$, thus $\dim \ker \iota = 0$ by rank nullity.
2. The claim proven above (equation (110)) alongside PBW establishes that a basis for U_k is given by $\left\{ x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n} \mid a_j \geq 0 \text{ and } \sum_j a_j \leq k \right\}$. So we may write

$$\dim U_k = \dim U_{k-1} + \# \left\{ (a_1, \ldots, a_n) \mid \sum_j a_j = k \right\} \tag{111}$$

Moreover we may proceed by induction since $\dim U_0 = \dim F = 1$.

TODO: prove claim